

Seunggeun Kim

✉ sgkim@utexas.edu [seunggeunkimkr.github.io](https://github.com/seunggeunkimkr) [in](#) Seunggeun Kim

Summary

I am a third-year Ph.D. student in Electrical and Computer Engineering at The University of Texas at Austin. My research focuses on sequence modeling with diffusion models and their real-world applications across diverse domains, including protein design, natural language processing, and code generation. I am also interested in integrating diffusion-based approaches with structured reasoning and large language models to enable more efficient and human-like reasoning.

Education

- PhD** **University of Texas at Austin**, Electrical and Computer Engineering Aug 2023 – Present
- Advisor: Adam Klivans
- BS** **Seoul National University**, Electrical and Computer Engineering Mar 2017 – Feb 2023
- GPA: 3.9/4.0 (summa Cum Laude)
 - 2 years absence to fulfill military duty

Publications

- [Fine-Tuning Masked Diffusion for Provable Self-Correction](#) [🔗](#), **ICML 2026, ICLR 2026 Workshop on MM Intelligence**
Seunggeun Kim*, Jaeyeon Kim*, Taekyun Lee*, David Z. Pan, Hyeji Kim, Sham Kakade, Sitan Chen
- [PPAAS: PVT and Pareto Aware Analog Sizing via Goal-conditioned Reinforcement Learning](#) [🔗](#), **ICCAD 2025**
Seunggeun Kim, Ziyi Wang, Sungyoung Lee, Youngmin Oh, Hanqing Zhu, Doyun Kim, David Z. Pan
- [Addressing Continuity and Expressivity Limitations in Differentiable Physical Optimization: A Case Study in Gate Sizing](#) [🔗](#), **ISEDA 2025**
 Yufan Du, Zizheng Guo, Yang Hsu, Zhili Xiong, **Seunggeun Kim**, David Z Pan, Runsheng Wang, Yibo Lin
- [DICE: Device-level Integrated Circuits Encoder with Graph Contrastive Pretraining](#) [🔗](#), **Under Review**
 Sungyoung Lee, Ziyi Wang, **Seunggeun Kim**, Taekyun Lee, Yao Lai, David Z. Pan
- [M3: Mamba-assisted Multi-Circuit Optimization via MBRL with Effective Scheduling](#) [🔗](#), **ICCAD 2025**
 Youngmin Oh, Jinje Park, **Seunggeun Kim**, Taejin Paik, David Pan, Bosun Hwang
- [LLM-enhanced Bayesian optimization for efficient analog layout constraint generation](#) [🔗](#), **arXiv preprint**
 Guojin Chen, Keren Zhu, **Seunggeun Kim**, Hanqing Zhu, Yao Lai, Bei Yu, David Z Pan

Experience

- Samsung AI Center**, Research Intern Suwon, Korea
 May 2024 – Jul 2024
- Mentor: Doyun Kim
 - Reinforcement Learning for PVT-aware analog sizing
- Seoul National University**, Research Intern Seoul, Korea
 Jan 2021 – Jul 2023
- Advisor: Jaeha Kim
 - Project lead of DSP tape-out with Samsung 14nm FinFET technology

Honors and Awards

- DAC Young Fellowship 2024
- UT Austin Engineering Fellowship 2023

Technologies

Programming Languages: Python, Verilog, C, C++, Java

Technologies: Pytorch, Tensorflow, Linux, GIT

Circuit Tools: Virtuoso, Spectre, Design Compiler, Vivado, XMODEL, NGSpice